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The Case for Increased Transatlantic Cooperation on Artificial Intelligence

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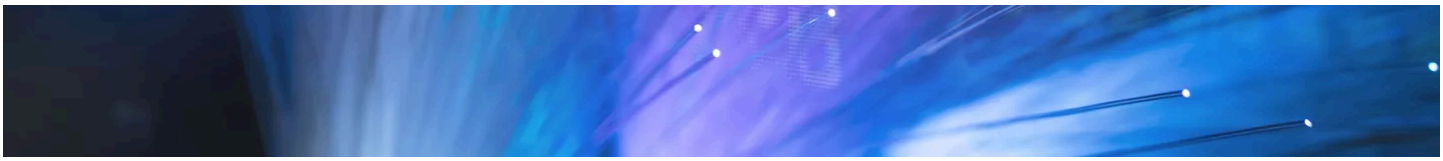

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Executive Summary

After being told in the wake of 9/11 that European and NATO allies pledged to fight Al Qaida alongside American troops, the then United States National Security Advisor Condoleezza Rice said "it was good to have friends in the world at a time like this."[1] Nicholas Burns, the then US Ambassador to NATO, has since reflected on the importance of the transatlantic alliance. Losing the relationship with NATO and members of the European Union, he believes, would lead the US to "lose our strongest anchor in a dangerous and complex world."[2]

The world has changed a lot since September 2001, however these relationships are no less important. Global terrorism is still a threat, but the rise of China and technological advancements have converged to create both new opportunities and new challenges. Artificial intelligence (AI) promises to help the world find a vaccine for Covid-19, add up to $15.7 trillion to the global economy, and improve militaries' ability to detect, defend, and deter against cyberattacks.[3] However, AI technologies could also provide adversaries and authoritarian governments with tools to increase censorship, automate disinformation. and engage in constant cyber or kinetic conflict.[4]

Despite all of these changes, the importance of a strong relationship between the United States and the European Union has been a constant. The transatlantic disagreements that have characterized the past few years—and have hampered a united front on emerging technologies like 5G and AI[5]—are not the first time US-EU relations have suffered, but they should not further divide allies that share common values.[6] Deepened US-EU cooperation across the entire AI ecosystem[7] is necessary to advance a more secure, safe, and prosperous world, but to do this the current level of AI-related coordination and partnership needs to be increased.

This report's purpose is twofold: first, to inform policymakers and researchers about the current state of transatlantic AI efforts; and second, to recommend specific areas where transatlantic AI collaboration should be strengthened. Based on a comprehensive study of over 260 documents and reports covering the period from December 1997 to June 2020, we proposes more than 16

recommendations to increase US-EU AI collaboration across the entire AI ecosystem, as well as 9 recommendations for AI cooperation in the healthcare, environmental sciences, and defense sectors. Greater transatlantic efforts are needed to prevent the advancement of an AI vision that is adversarial and harmful to the wellbeing of the United States, the European Union, and allies.

The Case for Transatlantic Cooperation

There are three critical, interconnected arguments for transatlantic cooperation to ensure AI innovation protects the security, values, and economic interests of the United States and the European Union.

1.Global Good: Transatlantic AI partnerships and cooperation encourages innovation and applications that enhance human welfare, strengthen the economies of the US and the EU, and advance global security.

2.Great Power Competition: US-EU leadership of like-minded nations is needed in this age of great power competition to tip the scales against efforts by authoritarian governments—particularly, China and Russia—to undermine democracies.

3.Shared Values: The US and the EU share fundamental values and would benefit from joint efforts to establish AI norms that would more effectively advance their common vision of AI and ripple throughout the global AI ecosystem.

Although the US consistently sounds the alarm bells around China's AI aspirations and the EU urges international efforts against AI that violates fundamental rights, increasingly noting China's actions with concern,[8] little concrete international action has taken place. The United States and the European Union's ongoing reassessment of their respective AI strategies and legislation[9] provides a window of opportunity to align and collaborate. Transatlantic AI cooperation is at a critical juncture and the United States and the European Union should seize this opportunity to take concrete actions.

The Current State

The United States and the European Union are separately assessing and updating their AI strategies. However, it is a myth to assume they are not collaborating at all to advance their AI-related goals. Transatlantic cooperation on AI norms, standards, research and development, and data sharing should increase, but the United States and the European Union can build upon an existing foundation for a stronger alliance.

United States: The United States views American leadership in AI as necessary to safeguard American values and maintain defense and economic superiority. Recognizing the need to develop a national AI approach and reclaim the AI R&D global leadership position from China, which had already surpassed the US in several research output metrics by 2016,[10] the Obama Administration developed an AI R&D prioritization in October 2016.[11] Building on this urgency, the Trump Administration has prioritized AI and established the American AI Initiative in February 2019.[12] This Initiative identified the need for a whole-of-government approach to prioritize AI R&D and deployment throughout the entire federal government. The Initiative also identifies the need to grow the US AI workforce, set national and global norms and standards, and work with industry and allies to promote an AI environment favorable to the United States.[13]

The United States' federal government has made key strategic and tactical changes to achieve these goals. Federal AI R&D and the American AI Initiative are coordinated by several committees and subcommittees within the Executive Office. President Trump pledged to more than double non-defense AI R&D to $2 billion by 2022.[14] Federal AI R&D, guided by the National AI R&D Strategic Plan, must now be reported annually for each federal entity.[15] The United States has taken a "light-touch" approach to regulation, fearing overly burdensome laws will stifle innovation. However, guidance is not completely absent. The Office of Management and Budget released a memo to guide Federal agencies as they develop regulatory and non-regulatory approaches to non-government applications of AI and the Department of Defense published five AI principles to guide AI design, deployment, and adoptions in defense.[16]

Obstacles to the US realizing its goal of global AI leadership exist, despite the government's prioritization of it. *Key obstacles include the need to bolster its private sector AI landscape; address regulatory or standards gaps to safeguard American values; repair the breakdown of funding and information sharing relationships between academia, industry, and government; grow its AI workforce; and further increase its federal AI R&D funding*.

European Union: The European Union, like the United States, intends to leverage AI's potential as a strategic and transformative technology.[17] However, the EU has positioned itself as a leader in trustworthy, human-centric, ethical, and values-based AI,[18] in comparison to the US government's emphasis on the need for AI innovation to protect American values, civil liberties, and privacy. The EU recognizes that it trails behind the US and China in terms of volume of investment and maturity of its tech industry.[19] Nonetheless, the EU believes it can capitalize on its underlying structural strengths (e.g., academic and innovation record) and on its values to compete globally and reaffirm its digital and technological sovereignty.[20] Starting with its 2018 *Communication: Artificial Intelligence for Europe*,[21, 22] the European Commission (EC) has launched a coordinated effort promoting AI.[23] Policies include increasing public and private investments from $5.6 billion to $22 billion annually;[24] coordinating research and innovation across Europe; devising ethical guidelines; fostering digital skills in its workforce; and promoting public and private sector adoption of AI.[25] To support and counsel these efforts, the EC has established the High-Level Expert Group on AI (AI

HLEG) comprising 52 experts who advise the Commission on policy and regulatory changes.

The European Union's Juncker[26] Commission (2014-2019) actively avoided regulating AI, causing the European Parliament to increase their efforts as a proactive voice in favor of stronger AI regulation. However, since the beginning of Ursula von der Leyen's tenure, the Commission has initiated efforts to adopt stronger regulation for AI applications (i.e., differentiating regulation of AI based on defined "high-risk" and "low-risk" sectors") and associated data spaces.[27,28] These legislative proposals and their associated discussions are planned to be completed by the end of 2020. During the strategic planning and budgeting process of its R&D programs, the EU committed to providing at least EUR10.7 billion[29] for AI-related research conducted between 2021 and 2027.[30] Despite these financial and political efforts, the EU still remains technologically dependent on the US and China and suffers from a lack of capital and private funding, decentralized and uncoordinated AI expertise, severe brain drain (including to the US), and slow adoption of AI programming in its education and public sectors.

Transatlantic Cooperation: Despite over 40 years of scientific relationships and projects between the United States and the European Union, AI-specific collaboration has been fraught with varying degrees of political and academic skepticism on both side of the Atlantic, notably within the European Commission and the governments of some Member States (e.g., France and Germany).[31] Such a dynamic is aggravated, in part, by the ever-deteriorating transatlantic relationship spurred by policy and trade disagreements, public spats, and increasing American isolationism. Despite such explicit omissions and stand-offs at the highest levels, transatlantic collaboration for AI does happen, most notably in various multilateral forums working on standards (e.g., ISO, IEC, IEEE, G7, G20) or on ethics and norms (e.g., OECD, GPAI[32]).[33] In recent months, however, interests and political support for greater transatlantic coordination on AI seems to be increasing. This trend was notably demonstrated by a visit from Lt. Gen. Jack Shanahan—then Director of the US Department of Defense's Joint Artificial Intelligence Center (JAIC)—to Brussels in January 2020 and a visit by the European Parliament's delegation to Washington D.C in February 2020. Both visits included discussions on AI with a variety of key stakeholders, such as NATO, representatives from the US Congress, State Department, Federal Transit Administration (FTA), Federal Bureau of Investigation (FBI), and Privacy and Civil Liberties Oversight Board (PCLOB).[34]

Transatlantic collaboration for AI-related research is taking place at varying levels although these projects are relatively ad hoc and materialize within existing scientific and technological research agreements and roadmaps. For instance, the current *Roadmap for US-EU Science & Technology* prioritizes four areas for transatlantic cooperation, most of which leverage AI (e.g., health, transportation, bioeconomy, marine and arctic research) or promote institutions that do (e.g., European Organization for Nuclear Research or CERN).[35,36] These collaborative links are supported and promoted through a variety of arrangements and initiatives, such as BILAT 4.0, EURAXES[37] or the European Network of Research and Innovation Centers and Hubs (ENRICH). In general, and despite challenges to systematically integrating US entities into European research programs, the US remains the leading non-EU ("third country") participant in Horizon 2020,[38] with over 60

participations and 1,200 partnerships.[39] US funding contributions to Horizon 2020 and participation in AI-related projects, however, is meager than its broader research involvement in Horizon 2020. For instance, US collaborative links with Horizon 2020 projects can only be found in 2% of AI-related projects, 12% of deep learning projects, and 4% of machine learning-related projects.[40] Accordingly, there is still plenty of room for improvement.[41]

We identified the healthcare, environmental sciences, and defense sectors as areas where the US and EU should prioritize their joint R&D efforts because of existing S&T cooperation, the importance of EU-US alignment for their joint security, and advancement of the "global good."

Health-related joint R&D is already a top priority within existing EU-US S&T collaboration and benefits from a reciprocal funding agreement between the US NIH and the EU. [42] Covid-19 and prioritization by both the US and the EU to develop AI applications for healthcare further the potential for stronger US-EU AI collaboration in this sector. The environmental sciences sector similarly benefits from preexisting strong transatlantic collaboration and increased focus for AI-related research. The EU's focus on developing a "European Green New Deal" will only raise the importance and quantity of European R&D in this field.[43] Greater defense-related AI cooperation is increasingly viewed as an imperative by the US, with the DOD Artificial Intelligence Strategy highlighting the need for international AI cooperation to "safeguard a free and open international order."[44] Recent positive visits and collaboration between the JAIC, NATO, and European allies indicate AI collaboration in the defense sector will grow.[45]

Challenges to Collaboration & Recommendations

Full US-EU collaboration faces five distinct, but interconnected obstacles (see Figure 1 below). At the highest level, the United States and European Union have some diverging geopolitical interests (section A) illustrated by: America's increasing isolationism, the European Union's rebalancing to become a third power, the European Union's resistance to adversarial discourse about China, and domestic political demands to focus resources on COVID-19 responses. Flowing out of the geopolitical landscape and political interests are three overarching considerations that are bolstered by differing beliefs about the role and size of government and can fuel US-EU disagreements around AI. These US national interests and EU common priorities are (section B): AI's impact on national security and economic interests, as well as the ethics and values that guide AI's development and use. Finally, aspects of the AI operating environment (sections C, D, and E), such as regulation and governance (including standards and operationalizing principles), funding, data spaces, hardware, and computing resources, provide tactical areas for disagreement or misalignment.

Figure 1: Overview of Challenges to Transatlantic Cooperation

These challenges are many but are not insurmountable. We recommend the following 16 actions to facilitate the full realization of US-EU AI collaboration. The complete rationale, recommendation, sub-recommendations, and additional considerations are found in the Challenges to Collaboration & Recommendations section.

Summary of Recommendations

A1 **Shift the Narrative from Adversarial to Collaborative:** The US should recognize the EU has its own, sometimes competing, interests that will not change through antagonistic demands alone. The EU should soften its stance on certain issues, recognizing both that adversarial rhetoric against the US may threaten collaboration and total technological sovereignty is unlikely.

A2 **Increase High-level Engagements:** High-level visits highlight the importance placed on US-EU collaboration, enhancing understanding and providing opportunities for greater alignment. Engagements should restart across the full interagency at the highest levels (e.g. Director, Secretary, and Commissioner level) once travel reopens or virtual substitutes are established.

A3 **Foster a Like-Minded Coalition:** Work together to build a larger coalition of nations that share their AI vision. Combined efforts will act as a force multiplier in strengthening alliances that serve as a counterweight to China and authoritarian regimes' efforts on the global stage.

B1 **Establish US-EU Dialogues:** Establish a Track 1 dialogue, potentially modelled after the Canada-EU Digital Dialogues, to strengthen relations, communicate points of agreement and disagreement, share best practices, and identify collaboration across the entire AI ecosystem.

Additional considerations: These dialogues should be inclusive, with not only government officials but also representatives from academia, business, and civil society present, and could be incorporated into existing Track 1 dialogues or an upcoming US-EU summit. Track 1.5 and 2 should supplement this formalized engagement.

Related recommendations: *Dialogue can enable and strengthen the execution of all other recommendations in this paper.*

B2 **Increase and Formalize AI-Related Joint R&D:** Increase joint R&D through various avenues (joint ventures, greater US involvement in Horizon 2020, formal R&D agreement, coordinating international private partnerships). Pool resources for greater impact and larger scale research on topics of importance for both the US and the EU.

Sub-recommendation: Research partnerships should span across the entire AI ecosystem, but we believe the healthcare, defense, and environmental sciences sectors should be prioritized, as well as joint efforts to operationalize principles, verification, and standards.

Related recommendations: C1, D3

B3 **Share Best Practices:** Facilitate coordination on priorities and findings, increase capacity building through information sharing and best practices. This can occur between the US and EU's various networks of Centers of Excellence[46], establishing a shared platform (like BILAT 4.0), or dialogues and networking events.

Sub-recommendation: To guide decisions and ensure AI R&D and use respects shared values, a focus on applied AI ethics and operationalizing principles should be at the table.

Related Recommendations: B2, B4, B5, C1, E2

B4 **Improve Workforce AI Literacy & Strengthen AI Talent:** Both the US and the EU need to increase AI literacy throughout their public sectors. Talent exchanges can increase AI alignment and capacity building of employees. US and EU government officials should coordinate on developing AI training. Academia should share best practices on developing AI curriculum for schools and universities.[47]

Sub-recommendation*:* Government, industry, and academia need AI experts and a workforce literate not just in the technical and political aspects of AI, but also its ethical implications.[48] Joint talent and education efforts should include a focus on AI ethics and trustworthy AI. The NSF's National AI Research Institute on Trustworthy AI should be included.

Related Recommendations: B1, B3

B5 **Counter Industrial Espionage & Nefarious Private Investment:** Intelligence and best practices sharing on identifying and preventing nefarious industry investments and IP theft, can help safeguard the US and EU's economic and security interests. The US and EU should explore options to establish and improve coordination of investment screening practices and information sharing between CFIUS and similar bodies.

Related Recommendations: B1, B3

C1 **Operationalize Principles, Verification Mechanisms, & Standards:** Joint efforts will help: 1) prevent inconsistencies that deepen disagreements, hamper commercial innovation, and preclude companies from entering both markets, and 2) increase their leverage in multilateral institutions to promote their shared AI vision. Track 1, 1.5, and 2 efforts should be pursued and include ethicists.

Related Recommendations: B1, B2, C2

C2 **Align Future AI-Related Regulation:** Track 1 and 1.5 dialogue on AI and data-related regulation can enable communication on rationales and concerns about proposals. More formalized coordination should occur early in the political process and can help prevent inconsistencies that hurt industry.

Related Recommendations: B1

D1 **Increase US Involvement in Horizon 2020 and Horizon Europe:** As a key EU research policy tool US participation is important and provides an existing avenue to increase joint research. The US should increase its funding contribution and the EU should designate the US as a third-country party.

Related Recommendations: B1, B2

D2 **Include the US in Future EU AI Mega-projects:** The EU should include US researches in its planned AI mega-project[49] to enable collaboration and research. If the US undertakes a similar project, EU researchers should be included.

Related Recommendations: B1, B2

D3 **Build on Existing US-EU Government Collaboration:** The NITRD AI R&D Interagency Working Group (WG) should conduct a survey of existing collaboration with European partners across all 24 federal agencies under its purview. The WG should identify and remediate impediments to collaboration

and further AI research that can build on existing research projects and relationships. The EU and its Member States should conduct a similar survey.

Related Recommendations: B2, B1

E1 **Enable Data Sharing for R&D:** As the EU is pursuing the creation of sector-specific data spaces and open datasets, the US and the EU should: 1) coordinate their data collection, sharing, use, re-use, access, and storage rules and standards, as well as 2) coordinate agreements to enable researchers have access to data spaces and relevant datasets.

Sub-recommendation: 1) Focusing on "low-risk" sectors (e.g., environmental sciences), as defined by the EU's AI HLEG, may be the most politically expedient as a starting point. 2) The USMCA's Digital Trade Chapter can be used as a model for an open data agreement.

Related Recommendations: B1, B2, C2, E2

E2 **Address Domestic Data Sharing Impediments to R&D:** Prioritize increasing domestic access to data spaces and datasets across government, industry, and academia. This will decrease challenges to transatlantic data sharing and joint R&D.

Related Recommendations: E1, B1, B2, B3

E3 **Identify Countries' Comparative Advantage in the Supply Chain:** Collaboratively identify countries, including members of a like-minded coalition, with a comparative advantage in producing and sourcing parts of AI-related hardware and resources.

Related Recommendations: A3

Recommendations for Healthcare, Environmental Sciences, & Defense Sectors

The healthcare, defense, and environmental sciences sectors should be prioritized in joint AI partnerships. There is existing US-EU collaboration in all three areas, they will continue to be of global importance, and alignment can be improved to the benefit both sides. We provide three specific recommendations for the healthcare sector, two recommendations for the environmental

sciences sector, and four recommendations for the defense sector. The full rationale, recommendation, sub-recommendations, and additional considerations can be found in the following sections: Healthcare Recommendations, Environmental Sciences Recommendations, and Defense Recommendations.

Healthcare

F1 **Focus AI Healthcare Research Related to Pandemics on Detection, Diagnosis, and Treatment:** COVID-19 highlights the necessity of international cooperation in addressing global pandemics. The US and the EU should explore, identify, and fund joint R&D on detecting (natural language processing, social media scanning), diagnosing (focus on few-shot and transfer learning[50, 51]), and treating (deep learning to generate novel drugs) pandemics.

Related Recommendations: B2, D3, D5, E1

F2 **Convene Experts and Policymakers to Address Legal and Ethical Obstacles:** Including healthcare experts, lawyers, ethicists, government officials, businesses, and other relevant experts to reflect on the challenges arising from AI applications in healthcare and COVID-19 and propose policy, technical, and legal/regulatory options to address obstacles.

Related Recommendations: B1, B3, C1, C2, E2

F3 **Provide Best Practices for Regulators:** AI applications pose new challenges to regulators working within legal frameworks that are outdated and ill-equipped to guide oversight of emerging technologies. The US and the EU should establish a mechanism for their regulators (FDA, EMA, other relevant government agencies) to share best practices on oversight of AI healthcare applications.

Related Recommendations: B1, B3, C1, C2

Environmental Sciences

G1 **The US Should Contribute Research to the EU's Green New Deal:** The EU is currently reviewing how the Green New Deal can help restart the economy, promote sustainable development, and increase innovation in emerging technologies like AI. The US and the EU should identify and fund joint research and development projects under this EC policy initiative.

Related Recommendations: B2, D1, D3, E1

G2 **Establish a Joint Funding Mechanism for Environmental Sciences AI R&D:** The US and the EU should explore funding for environmental sciences research, either as part of Horizon 2020, Horizon Europe, or as a separate agreement. This can be modelled after the reciprocity agreement between the EC and the US National Institutes of Health, National Science Foundation, and the Bill and Melinda Gates Foundation for health research.[52] Climate-related research, should be prioritized.

Related Recommendations*: D1*

Defense

H1 **Shift the Narrative Away from Lethal Autonomous Weapons (LAWS):** The development of lethal autonomous weapons is causing tension that may prohibit substantive discussion around new areas for collaboration. The US and the EU should shift the conversation away from potential disagreement around LAWS and towards shared defense priorities such as ensuring military interoperability separate from autonomous weapons development.

Related Recommendation: A2

H2 **Relax Restrictions on Third-Country Funding, IP Rights:** The EU should consider reviewing and changing its EDF and PESCO regulations, allowing non-EU companies to receive funds and maintain IP rights in certain collaborative research projects. The EU and member state governments should consider not replicating these restrictions in other defense-related R&D mecha-

nisms and collaborative efforts with the US.

H3 **The US and the EU should Strengthen their Defense-Related AI Talent:** The US and the EU should work together to pool their defense AI talent to address workforce gaps. This could include defense-related talent exchanges, talent exchanges/secondments into industry to strengthen AI literacy and skills, coordination on AI training and educational programs, and sharing of associated best practices.

H4 **Increase Efforts to Share Defense-Related Data:** The US and the EU should remove obstacles to sharing defense and intelligence-related data where appropriate. They should also fund and prioritize addressing challenges to data sharing through NATO. Efforts should begin immediately as data sharing and governance projects are often lengthy undertakings. Projects focused on achieving a specific, concrete goal should lay the groundwork for larger-scale, general efforts.[53]

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Executive Summary Footnotes:

1 This recounting is from Nicholas Burns. See: https://www.belfercenter.org/publication/experts-weigh-transatlantic-relationship 2 "Experts Weigh in On Transatlantic Relationship" (Harvard Kennedy School Belfer Center for Science and International Affairs, April 24, 2018), https://www.belfercenter.org/publication/experts-weigh-transatlantic-relationship. 3 Stevent Rosenbush, "In Race to Treat Coronavirus, AI Is Seen as Key" (The Wall Street Journal, March 16, 2020), https://www.wsj.com/articles/in-race-to-treat-coronavirus-ai-is-seen-as-key-11584351000?tpl=artificialintelligence; "AI Analysis: Sizing the Prize" (PWC, 2017), https://www.pwc.com/gx/en/issues/analytics/assets/pwc-ai-analysis-sizing-the-prize-report.pdf; "Summary of the 2018 Department of Defense Artificial Intelligence Strategy: Harnessing AI to Advance Our Security and Prosperity" (Department of Defense, February 2019), https://media.defense.gov/2019/Feb/12/2002088963/-1/-1/1/SUMMARY-OF-DOD-AI-STRATEGY.PDF. 4 Jason Skowronski, "Trolls and Bots Are Disrupting Social Media - Here's How AI Can Stop Them" (Medium, July 30, 2019), https://towardsdatascience.com/trolls-and-bots-are-disrupting-social-media-heres-how-ai-can-stop-them-d9b969336a06; Miles Brundage et al., "The Malicious Use of Artificial Intelligence: Forecasting, Prevention, and Mitigation," February 2018, https://img1.wsimg.com/blobby/go/3d82daa4-97fe-4096-9c6b-376b92c619de/downloads/MaliciousUseofAI.pdf?ver=1553030594217; Cade Metz and Scott Blumenthal, "How A.I. Could Be Weaponized to Spread Disinformation" (The New York Times, June 7, 2019), https://www.nytimes.com/interactive/2019/06/07/technology/ai-text-disinformation.html; "Welcome to the New Era of Chinese Government Disinformation" (The Diplomat, May 11, 2020), https://thediplomat.com/2020/05/welcome-to-the-new-era-of-chinese-government-disinformation/; Paul Mozur, "Inside China's Dystopian Dreams: A.I., Shame and Lots of Cameras" (The New York Times, July 8, 2018), https://www.nytimes.com/2018/07/08/business/china-surveillance-technology.html. 5 Erik Brattberg and Philippe Le Corre, "Huawei and Europe's 5G Conundrum" (Carnegie Endowment for International Peace, December 27, 2018), https://carnegieendowment.org/2018/12/27/huawei-and-europe-s-5g-conundrum-pub-78045. 6 Nicholas Burns Ambassador (ret.), "The Transatlantic Relationship in Crisis" (Madrid, Spain, July 6, 2019), https://www.belfercenter.org/publication/transatlantic-relationship-crisis. 7 We define the AI ecosystem to include not only the software, algorithms, systems, and data sets required for general AI, machine-leaning, and deep learning systems, but also the associated hardware, computing resources, "laws, funding, institutions, policies, talent, intellectual property protection, [and] supply chains" that enable the research, development, and use of AI applications. "Interim Report" (National Security Commission on Artificial Intelligence, November 2019). 8 The EC's *Communication: Artificial Intelligence for Europe* (2018) only explicitly mentions China's investment capacity in AI as a concern. However, the EC's High-Level Expert Group on AI's (AI HLEG) *Policy and Investment Recommendations for Trustworthy* (2019) report underlined the risks generated by "identifying and tracking individuals with AI," "covert AI systems," and "AI enabled citizen scoring in violation of fundamental rights," which underpins China's social credit system. 9 The EU is planning to propose AI-related legislation by the end of 2020. 10 Ajay Agrawal, Joshua Gans, and Avi Goldfarb, "The

Obama Administration's Roadmap for AI Policy" (Harvard Business Review, December 21, 2016), https://hbr.org/2016/12/the-obama-administrations-roadmap-for-ai-policy 11 Select Committee on Artificial Intelligence of the National Science & Technology Council, "The National Artificial Intelligence Research and Development Strategic Plan: 2019 Update" (Executive Office of the President of the United States, June 2019), https://www.nitrd.gov/pubs/National-AI-RD-Strategy-2019.pdf 12 The American AI Initiative was founded through the Executive Order on Maintaining American Leadership in Artificial Intelligence. 13 "Executive Order 13859: Maintaining American Leadership in Artificial Intelligence" (Executive Office of the President, February 11, 2019). 14 "President Trump's FY 2021 Budget Commits to Double Investments in Key Industries of the Future" (The White House, February 11, 2020), https://www.whitehouse.gov/briefings-statements/president-trumps-fy-2021-budget-commits-double-investments-key-industries-future/ 15 Subcommittee on Networking & Information Technology Research & Development Committee on Science & Technology Enterprise of the National Science & Technology Council, "The Networking & Information Technology Research and Development Program: Supplement to the President's FY 2020 Budget" (Executive Office of the President of the United States, September 2019). 16 Russell T. Vought, "Memorandum for the Heads of Executive Departments and Agencies: Guidance for Regulation of Artificial Intelligence Applications" (Office of Management and Budget, January 13, 2020), https://www.whitehouse.gov/wp-content/uploads/2020/01/Draft-OMB-Memo-on-Regulation-of-AI-1-7-19.pdf 17 European Commission, "Artificial Intelligence for Europe" (European Commission, April 25, 2018), https://ec.europa.eu/transparency/regdoc/rep/1/2018/EN/COM-2018-237-F1-EN-MAIN-PART-1.PDF 18 High level expert group on AI, "POLICY AND INVESTMENT RECOMMENDATIONS FOR TRUSTWORTHY AI" (Brussels: European Commission, June 29, 2019); European Commission, "Building Trust in Human-Centric Artificial Intelligence" (European Commission, August 4, 2019). 19 M Craglia et al., "Artificial Intelligence A European Perspective" (Luxembourg: Joint Research Center - European Commission, 2018); Charlotte Stix, "A Survey of the European Union's Artificial Intelligence Ecosystem" (European Commission, February 1, 2019), https://doi.org/10.13140/RG.2.2.30791.65447 20 European Commission, "Artificial Intelligence for Europe," April 25, 2018. 21 A "Communication" is a high-level policy paper which is often part of the standard policy making procedure of the EC. 22 European Commission, "Artificial Intelligence for Europe," April 25, 2018. 23 European Commission; European Commission, "Coordinated Plan on Artificial Intelligence" (European Commission, July 12, 2018); Stix, "A Survey of the European Union's Artificial Intelligence Ecosystem"; European Commission, "Building Trust in Human-Centric Artificial Intelligence." 24 Digital Single Market, "Factsheet: Artificial Intelligence for Europe," 2019; European Commission, "Coordinated Plan on Artificial Intelligence"; Stix, "A Survey of the European Union's Artificial Intelligence Ecosystem"; European Commission, "Building Trust in Human-Centric Artificial Intelligence." 25 European Commission, "Artificial Intelligence for Europe," April 25, 2018; European Commission, "Coordinated Plan on Artificial Intelligence"; Stix, "A Survey of the European Union's Artificial Intelligence Ecosystem"; European Commission, "Building Trust in Human-Centric Artificial Intelligence." 26 Jean-Claude Juncker was the former president of the European Commission, which is the executive body of the European Union. 27 According to the EC, European data spaces should entail the clarification and harmonization of data governance models and practices, while setting up the necessary infrastructure to foster the exchange of quality and interoperable data in the respective sectors (e.g., public sector, health or banking sectors) "Annex: All Reports of the Workshops on 'Common European Data Spaces,'" (European Commission, July to November 2019). 28 Mark Scott, "What's Driving Europe's New Aggressive Stance on Tech" (Politico, October 27, 2019), https://www.politico.eu/article/europe-digital-technological-sovereignty-facebook-google-amazon-ursula-von-der-leyen/; Tyson Barker, "Europe Can't Win the Tech War It Just Started" (Foreign Policy, January 16, 2020), https://foreignpolicy.com/2020/01/16/europe-technology-sovereignty-von-der-leyen/; European Commission, "Annex: All Reports of the Workshops on "Common European Data Spaces"" (European Commission, Directorate General for Communications Networks, Content and Technolgy, November 2019), https://ec.europa.eu/digital-single-market/en/news/stakeholders-dialogue-common-european-data-spaces 29 This is roughly equivalent to $10.63 billion, depending

on the exchange rate. 30 "Horizon Europe - the next Research and Innovation Framework Programme," ec.europa, n.d., https://ec.europa.eu/info/horizon-europe-next-research-and-innovation-framework-programme_en; European Commission, "Artificial Intelligence," European Commission, July 12, 2018, https://ec.europa.eu/commission/news/artificial-intelligence-2018-dec-07_en. 31 Barker, "Europe Can't Win the Tech War It Just Started"; Thomas Metzinger, Professor of Theoretical Philosophy at the Johannes Gutenberg University of Mainz, April 14, 2020; Ulrike Esther Franke, Policy Fellow at European Council on Foreign Relations, June 24, 2020. 32 The Global Partnership on AI (GPAI), originally proposed in 2018 by France and Canada as the International Panel on Artificial Intelligence, was formally launched in May 2020 by the G-7 science and technology ministers. The US was originally hesitant to join but has now thrown its support behind the initiative to counter the threat posed by China and authoritarian regimes. 33 Eanna Kelly, "US Joins Global AI Group, Citing Technology Threat from China" (Science Business, May 29, 2020), https://sciencebusiness.net/international-news/us-joins-global-ai-group-citing-technology-threat-china; Dave Nyczepir, "The US Joined a Global AI Partnership for Coronavirus Recovery to Stick It to China" (FedScoop, May 28, 2020), https://www.fedscoop.com/us-global-ai-partnership-china/; The White House Office of Science and Technology Policy, "Summary of the 2018 White House Summit on Artificial Intelligence For American Industry," May 10, 2018, https://www.whitehouse.gov/wp-content/uploads/2018/05/Summary-Report-of-White-House-AI-Summit.pdf?latest#page=13. 34 Katie Malone, "JAIC Director: Future of Defense AI Relies on Global Collaboration," MeriTalk, January 17, 2020, https://www.meritalk.com/articles/jaic-director-future-of-defense-ai-relies-on-global-collaboration/; European Parliament Liason Office in Washington DC, "EU-US Relations in Data Protection, AI and Security: MEPs Conclude Visit to US," European Parliament, February 28, 2020, https://www.europarl.europa.eu/unitedstates/en/eplo-news/eu-us-relations-in-data-protection-ai-and-security-meps-conclude-visit-to-us; Department of State, "Press Briefing with Air Force Lt. Gen. John Shanahan, Director of Joint Artificial Intelligence Center, US Department of Defense," state.gov, January 15, 2020, https://www.state.gov/120728-2/. 35 CERN is one of the largest European research organization that operates the largest particle physics laboratory in the world. While not a full member, the US has an observer status and collaborates on a number of projects. 36 European Commission, "Roadmap for EU USA S&T Cooperation" (European Commission, 2017), https://ec.europa.eu/research/iscp/pdf/policy/us%20clean_roadmap_2017.pdf. 37 BILAT 4.0 was a coordination and support project under Horizon 2020, which aimed to enhance and develop science, technology, and innovation (STI) partnerships between the EU and the US. EURAXES is a networking platform for STI that aims to facilitate the mobility of researchers across the world, including in the US. 38 Horizon 2020 is a 6-year (2014-2020) EU research framework. It is used as a policy instrument to implement and fund high-level EU policy initiatives. Its estimated budget is approximately EUR 80 billion. 39 Taken out of the Community Research and Development Information Service (CORDIS), which is the EC's primary source of results from the projects funded by the EU's framework programs for research and innovation. 40 Ibid. 41 This is based off original research by the authors. See *Current State: Transatlantic Cooperation* section for more explanation. 42 "Horizon 2020: International Cooperation Opportunities in the WOrk PRogramme 2016-2017" (European Commission, Directorate-General for Research and Innovation, 2016), http://ec.europa.eu/research/iscp/pdf/iscp_wp_2016_17.pdf; European Commission, "Roadmap for EU USA S&T Cooperation." 43 Metzinger, Professor of Theoretical Philosophy at the Johannes Gutenberg University of Mainz. 44 "Summary of the 2018 Department of Defense Artificial Intelligence Strategy: Harnessing AI to Advance Our Security and Prosperity" (Department of Defense, February 2019), https://media.defense.gov/2019/Feb/12/2002088963/-1/-1/1/SUMMARY-OF-DOD-AI-STRATEGY.PDF; "DOD Unveils Its Artificial Intelligence Strategy" (Defense.gov, February 12, 2019), https://www.defense.gov/Explore/News/Article/Article/1755942/dod-unveils-its-artificial-intelligence-strategy/. 45 Malone, "JAIC Director: Future of Defense AI Relies on Global Collaboration"; Nand Mulchandani, Acting Director of the U.S. Department of Defense Joint Artificial Intelligence Center, May 22, 2020. 46 The United States' GSA has an AI Center of Excellence focused on AI

applications within government and NSF will establish National AI Research Institutes in partnership with academia and other Departments. The EU is in the process of establishing various networks of Centers of Excellence and European Digital Innovation Hubs. 47 The EU is currently working on Artificial Intelligence and Analytics in their Digital Education Action Plan; the US could learn from this commission's work and provide beneficial input. 48 Metzinger, Professor of Theoretical Philosophy at the Johannes Gutenberg University of Mainz. 49 European researchers have called for a CERN for AI, European Lab for Learning and Intelligent Systems (ELLIS), and the Confederation of Laboratories for Artificial Intelligence Research in Europe (CLAIRE). 50 Few-shot learning refers to machine learning algorithms that learn from a few data points or examples. One-shot and zero-shot learning refers to AI that can use one or zero data points respectively. Few-shot learning has gained momentum as a field of research particularly after Google DeepMind published Matching Networks for One Shot Learning in 2016. Transfer learning refers to AI that, although trained to do one task, can adapt to do a similar task. 51 Rob Toews, "Questioning the Long-Term Importance of Big Data in AI" (Forbes, November 4, 2019), https://www.forbes.com/sites/robtoews/2019/11/04/questioning-the-long-term-importance-of-big-data-in-ai/#6b08938a2177; Will Douglas Heaven, "AI Could Help with the Next Pandemic--But Not With This One" (MIT Technology Review, March 12, 2020), https://www.technologyreview.com/s/615351/ai-could-help-with-the-next-pandemicbut-not-with-this-one/. 52 BILAT 4.0, "EU-US STI COOPERATION PATTERNS—STATUS QUO" (European Union's Horizon 2020 Research and Innovation Programme, 2016), 0, https://www.euussciencetechnology.eu/assets/content/Deliverables/BILAT_USA_4.0_Deliverable_2.1_Report_on_Status_quo_and_EU-US_STI_Cooperation_patterns_v1_forweb.pdf. 53 Informed by interviews with Nand Mulchandani, Acting Director of the U.S. Department of Defense Joint Artificial Intelligence Center, and Ulrike Esther Franke, Policy Fellow at European Council on Foreign Relations.

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